# Inverse Reinforcement Learning for Mean-field Games with Average Reward Criterion

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# Mean-Field Game

An Mean-Field Game is defined by a tuple: $(\mathcal{X},\ \mathcal{A},\ p,\ r)$ where,

- $\mathcal{X}$ and $\mathcal{A}$ are finite "state" and "action" spaces, respectively.
- $p : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$ is transition probability, determines how the next state evolves.
- $r : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \to \mathbb{R}$ is one-stage reward function, specifies the immediate reward recieved after each state.

# Mean-Field Term $\mu$

In a population of $N$ agents with states $(x_1, x_2, \ldots, x_N)$, the empirical distribution $\mu^{(N)}$ converges (in law) to a deterministic distribution $\mu \in \mathcal{P}(\mathcal{X})$, known as the *mean-field term*.

$$\mu^{(N)} := \frac{1}{N} \sum_{i=1}^{N} \delta_{x_i} \xrightarrow{N \to \infty} \mu \in \mathcal{P}(\mathcal{X})$$

represents the collective behavior of all agents.

We focus on the stationary setting, where the population distribution stabilizes and remains constant over time

# Dynamics

An agent follows a *stationary Markov policy*, that is, a stochastic kernel:

$$\pi : \mathcal{X} \to \mathcal{P}(\mathcal{A}), \quad x \mapsto \pi(\cdot \mid x)$$

meaning that given state $x_t = x$, the action $a_t$ is sampled from $\pi(\cdot \mid x)$.
The state evolves as:

$$x_{t+1} \sim p(\,\cdot \mid x_t,\ a_t,\ \mu)$$

Let $\Pi$ denote all such policies.

## Definition

For a fixed $\mu$, the **average reward** of a policy $\pi \in \Pi$ is given by

$$V_\mu(\pi, \mu_0) = \limsup_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^\pi\Big[r(x_t,\ a_t,\ \mu)\Big], \quad \textit{where} \quad x_t \sim \mu_0$$

# Mean-Field Equilibrium

Define the following set-valued mappings:

$$\Phi\colon \mathcal{P}(\mathcal{X}) \to 2^{\Pi}, \quad \mu \mapsto \left\{ \hat{\pi} \in \Pi \;\middle|\; V_\mu(\hat{\pi}, \mu) = \sup_{\pi \in \Pi} V_\mu(\pi, \mu) \right\}$$

$$\Lambda\colon \Pi \to 2^{\mathcal{P}(\mathcal{X})}, \quad \pi \mapsto \left\{ \mu_\pi \in 2^{\mathcal{P}(\mathcal{X})} \;\middle|\; \mu_\pi(\,\cdot\,) = \sum_{x \in \mathcal{X}} p(\,\cdot \mid x, a, \mu_\pi)\, \pi(a \mid x)\, \mu_\pi(x) \right\}$$

Namely, $\Phi(\mu)$ represents the set of optimal policies for a given state measure $\mu$. Similarly, $\Lambda(\pi)$ represents the set of stationary state distributions induced by policy $\pi$.

**Definition**

A pair $(\pi_*, \mu_*) \in \Pi \times \mathcal{P}(\mathcal{X})$ is called a *mean-field equilibrium* (MFE) if $\pi_* \in \Phi(\mu_*)$ and $\mu_* \in \Lambda(\pi_*)$.

# Forward vs Inverse in MFGs

**Forward MFG (RL)**

| Reward $r(x, a, \mu)$ | $\rightarrow$ | Optimization<br>$\pi^* \in \Phi(\mu)$ | $\rightarrow$ | Equilibrium<br>$(\pi^*, \mu^*)$ |
|---|---|---|---|---|

**Inverse MFG (IRL)**

| Expert Behavior<br>$(\pi_E, \mu_E)$ | $\rightarrow$ | Feature Stats<br>$\langle f \rangle_{\pi_E, \mu_E}$ | $\rightarrow$ | Max Entropy<br>Optimization | $\rightarrow$ | Recovered policy<br>$\pi^*$ |
|---|---|---|---|---|---|---|

# Inverse RL

**Assumption 1:** The reward is a linear combination of features that depend on the state, action, and mean-field term:

$$\mathcal{R} := \left\{ r(x, a, \mu) = \langle \theta, f(x, a, \mu) \rangle \;\middle|\; \theta \in \mathbb{R}^k,\ f : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \to \mathbb{R}^k \right\}$$

where $f(x, a, \mu) \in \mathbb{R}^k$ is the feature vector, which is assumed to be known and fixed.

**Assumption 2:** $\forall \pi \in \Pi,\ \forall \mu \in \mathcal{P}(\mathcal{X}) :\ \{x_t\}_{t \geq 0}$ is ergodic[1].
That is, the Markov chain admits a unique invariant distribution $\mu_\pi \in \mathcal{P}(\mathcal{X})$, and for any function $h : \mathcal{X} \to \mathbb{R}$,

$$\frac{1}{T} \sum_{t=0}^{T-1} h(x_t) \xrightarrow[T \to \infty]{\text{a.s.}} \sum_{x \in \mathcal{X}} h(x)\, \mu_\pi(x).$$

[1] P. Meyn and R. L. Tweedie, *Markov Chains and Stochastic Stability*, 2nd ed., Cambridge University Press, 2009, Theorem 13.3.3.

In IRL environment, under some mean-field equilibrium $(\pi_E, \mu_E)$ we have trajectories

$$\mathcal{Z} = \left\{ \left( x_t^i,\, a_t^i \right)_{t \geq 0} \right\}_{i=1}^{d}$$

Since $\mu_E$ is an invariant probability distribution under the policy $\pi_E$, and the mean-field term appears in the state dynamics as $\mu_E$, using $h(x) = \mathbf{1}_{\{x^i = x_0\}}$ in Assumption 2, ergodic theorem says:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbf{1}_{\{x_t^i = x_0\}} \xrightarrow[T \to \infty]{\text{a.s.}} \sum_{x \in \mathcal{X}} \mathbf{1}_{\{x^i = x_0\}} \, \mu_E(x) = \mu_E(x_0)$$

Hence we have,

$$\lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \left( \frac{1}{d} \sum_{i=1}^{d} \mathbf{1}_{\{x_t^i = x\}} \right) = \mu_E(x), \quad \forall x \in \mathcal{X}.$$

# Feature Expectation Vector

## Definition

**Expected feature vector** under the mean-field equilibrium $(\pi_E, \mu_E)$ for $x_0 \sim \mu_E$ is defined as,

$$\langle f \rangle_{\pi_E,\mu_E} := \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi_E,\mu_E} \left[ f(x_t, a_t, \mu_E) \right]. \tag{1}$$

If $d$ is large enough, we can obtain an estimate for the feature expectation vector as follows:

$$\frac{1}{d} \sum_{i=1}^{d} \left( \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} f(x_t^i, a_t^i, \mu_E) \right) \simeq \langle f \rangle_{\pi_E,\mu_E}$$

# Maximum Causal Entropy

## Definition

**Average-reward causal entropy** $H(\pi)$ of the policy $\pi \in \Pi$ is defined as follows

$$H(\pi) := \lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi, \mu_E} \left[ -\log \pi(a_t \,|\, x_t) \right].$$

Maximizing causal entropy means we choose the least 'biased' (most random) policy subject to matching constraints

# $(\mathbf{OPT})_1$

$$
\begin{array}{lll}
(\mathbf{OPT})_1 & \text{maximize}_{\pi \in \Pi} & H(\pi) \\
& \text{subject to} & \pi(a \mid x) \geq 0 \ \forall (x,a) \in \mathcal{X} \times \mathcal{A} \\
& & \sum_{a \in \mathcal{A}} \pi(a \mid x) = 1 \ \forall x \in \mathcal{X} \\
& & \mu_E(x) = \sum_{(a,y) \in \mathcal{A} \times \mathcal{X}} p(x \mid y, a, \mu_E)\, \pi(a \mid y)\, \mu_E(y) \ \forall x \in \mathcal{X} \\
& & \lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi, \mu_E}[f(x_t, a_t, \mu_E)] = \langle f \rangle_{\pi_E, \mu_E}
\end{array}
$$

The first two constraints ensure that $\pi \in \Pi$. The third and fourth constraints guarantee that the pair $(\pi, \mu_E)$ forms a MFE.

# Why $\mathbf{(OPT)_1}$ yields an MFE?

Let $\pi^*$ be a solution of $\mathbf{(OPT)_1}$, then:

- From the third constraint, $\mu_E$ is the stationary distribution under $\pi^*$, i.e., $\mu_E \in \Lambda(\pi^*)$.
- From the fourth constraint, $\pi^*$ is an optimal policy for $\mu_E$, i.e., $\pi^* \in \Phi(\mu_E)$.

  Since $\pi^*$ matches expert features and $\pi_E \in \Phi(\mu_E)$, we have

$$J_{\mu_E}(\pi^*, \mu_E) = \langle \theta_E, \langle f \rangle_{\pi_E, \mu_E} \rangle = J_{\mu_E}(\pi_E, \mu_E) = \sup_{\pi \in \Pi} J_{\mu_E}(\pi, \mu_E)$$

Thus, solving $\mathbf{(OPT)_1}$, leads to a mean-field equilibrium.

# Convex Reformulation of **(OPT)$_1$**

## Definition

We define the **state-action occupation measure** $\nu_\pi$ for any policy $\pi \in \Pi$ as

$$\nu_\pi(x,a) := \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{(x_t,a_t)=(x,a)\}}\right],$$

and corresponding **state occupation measure** as

$$\nu_\pi^{\mathcal{X}}(x) := \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{x_t=x\}}\right] = \sum_{a\in\mathcal{A}} \nu_\pi(x,a).$$

Notice that $\nu_\pi$ is a probability measure for any policy $\pi \in \Pi$.

# Stationarity Under Occupation Measures

## Lemma 1 (Stationarity)

If $\pi \in \Pi$ is feasible for $\textbf{(OPT)}_{\mathbf{1}}$, then $\nu_\pi^{\mathcal{X}} = \mu_E$. Furthermore, for all $x \in \mathcal{X}$, we have

$$\nu_\pi^{\mathcal{X}}(x) = \sum_{(y,a)} p(x \mid y, a, \mu_E)\, \nu_\pi(y, a).$$

*Proof.* Since $\pi$ is feasible for $\textbf{(OPT)}_{\mathbf{1}}$, it satisfies the constraint that $\mu_E$ is the stationary distribution under policy $\pi$. From the definition of $\nu_\pi^{\mathcal{X}}$, we have:

$$\nu_\pi^{\mathcal{X}}(x) = \lim_{T\to\infty} \sum_{t=0}^{T-1} \sum_{a\in\mathcal{A}} \mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{(x_t,a_t)=(x,a)\}}\right] = \lim_{T\to\infty} \sum_{t=0}^{T-1} \mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{x_t=x\}}\right].$$

By stationarity and the ergodic theorem, this limit equals to $\mu_E(x)$. This completes the first part of the proof...

For the second part, observe that:

$$\begin{aligned}\sum_{(y,a)\in\mathcal{X}\times\mathcal{A}} p(x \mid y,a,\mu_E)\,\nu_\pi(y,a) &= \lim_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\sum_{(y,a)}\mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{(x_t,a_t)=(y,a)\}}\right]p(x \mid y,a,\mu_E)\\ &= \lim_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}^{\pi,\mu_E}\left[p(x \mid x_t,a_t,\mu_E)\right]\\ &= \lim_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}^{\pi,\mu_E}[\mathbf{1}_{\{x_{t+1}=x\}}] = \mu_E(x).\end{aligned}$$

Last line follows from ergodicity and the law of total expectation. i.e,

$$\mathbb{E}^{\pi,\mu_E}\left[p(x \mid x_t,a_t,\mu_E)\right] = \mathbb{E}^{\pi,\mu_E}\left[\mathbb{E}\left[\mathbf{1}_{\{x_{t+1}=x\}} \mid x_t,a_t\right]\right] = \mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{x_{t+1}=x\}}\right].$$

Using the first part ($\nu_\pi^{\mathcal{X}} \equiv \mu_E$), we conclude the second part of the proof ■

# Rewrite the Entropy

## Lemma 2 (Entropy)

If $\pi \in \Pi$ is a feasible point for $\mathbf{(OPT)_1}$, then $H(\pi)$ can be written as

$$H(\pi) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} -\log\left(\frac{\nu_\pi(x,a)}{\mu_E(x)}\right)\nu_\pi(x,a).$$

*Proof.* We have

$$H(\pi) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} -\log \pi(a \,|\, x)\nu_\pi(x,a).$$

Here, the occupancy measure $\nu_\pi$ can be decomposed as

$$\nu_\pi(x,a) = \lim_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}^{\pi,\mu_E}\left[\mathbf{1}_{\{(x_t,a_t)=(x,a)\}}\right] = \pi(a \,|\, x)\nu_\pi^{\mathcal{X}}(x).$$

Substituting $\pi(a \mid x)$ and using previous Lemma concludes proof. ∎

# Rewrite Feature Constraint

## Lemma 3 (Feature Matching)

The average-reward feature expectation vector can be written as

$$\langle f \rangle_{\pi,\mu_E} = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} f(x,a,\mu_E)\nu_\pi(x,a)$$

*Proof.* By definition, the average-reward feature expectation vector under policy $\pi$ and invariant distribution $\mu_E$ is given by

$$\langle f \rangle_{\pi,\mu_E} := \lim_{T\to\infty} \sum_{t=0}^{T-1} \mathbb{E}^{\pi,\mu_E}[f(x_t,a_t,\mu_E)].$$

Using the identity

$$f(x_t,a_t,\mu_E) = \sum_{(x,a)} f(x,a,\mu_E)\mathbf{1}_{\{(x_t,a_t)=(x,a)\}}$$

and linearity of expectation, the limit distributes over the sum, that concludes proof. ∎

# $\textbf{(OPT)}_2$ : Occupation Measure-Based Form

In view of above results, we reformulate the problem with respect to the set of occupation measures instead of policies in $\Pi$.

$$\begin{aligned}
\textbf{(OPT)}_2 \ \text{maximize}_{\nu \in \mathbb{R}^{\mathcal{X}\times\mathcal{A}}} \quad & \textstyle\sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} -\log\left(\frac{\nu(x,a)}{\mu_E(x)}\right)\, \nu(x,a) \\
\text{subject to} \quad & \textstyle\sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} f(x,a,\mu_E)\,\nu(x,a) = \langle f\rangle_{\pi_E,\mu_E} \\
& \mu_E(x) = \textstyle\sum_{(y,a)\in\mathcal{X}\times\mathcal{A}} p(x \mid y,a,\mu_E)\,\nu(y,a) \ \forall x \in \mathcal{X} \\
& \nu^{\mathcal{X}}(x) = \mu_E(x) \ \forall x \in \mathcal{X} \\
& \nu(x,a) \geq 0 \ \forall (x,a) \in \mathcal{X}\times\mathcal{A}.
\end{aligned}$$

Where $\nu^{\mathcal{X}}(x) := \sum_{a\in\mathcal{A}} \nu(x,a)$.

# Convexity of **(OPT)$_2$**

**Lemma 4**

**(OPT)$_2$** is a convex optimization problem.

*Proof.* We can rewrite the objective function as

$$\sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} -\log\left(\nu(x,a)\right)\nu(x,a) + \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} \log\left(\mu_E(x)\right)\nu(x,a).$$

The first term of the above is the entropy of the distribution $\nu$, which is known to be strongly concave in $\nu$.[2] Moreover, the second term is linear in $\nu$. Therefore, the objective function is strongly concave.

Also note that all the constraints in **(OPT)$_2$** are linear in $\nu$. ■

[2] F. Alajaji and P.-N. Chen, *An Introduction to Single-User Information Theory*, Springer, 2018, p. 37.

# Remark

We can express the objective function **$(\mathrm{OPT})_2$** in a Min-Max form by introducing penalty variables $\boldsymbol{\alpha} \in \mathbb{R}^k$, $\boldsymbol{\beta}, \boldsymbol{\theta} \in \mathbb{R}^{\mathcal{X}}$ to enforce the constraints of **$(\mathrm{OPT})_2$**.

$$
\begin{aligned}
\textbf{(OPT)}_2 = \max_{\nu \in \mathcal{P}(\mathcal{X}\times\mathcal{A})} \min_{\substack{\boldsymbol{\alpha}\in\mathbb{R}^k \\ \boldsymbol{\beta},\boldsymbol{\theta}\in\mathbb{R}^{\mathcal{X}}}} \Bigg\{ & \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} -\log\left(\frac{\nu(x,a)}{\mu_E(x)}\right)\nu(x,a) \\
& + \langle \boldsymbol{\alpha}, \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} f(x,a,\mu_E)\nu(x,a) - \langle f\rangle_{\pi_E,\mu_E} \rangle \\
& + \sum_{z\in\mathcal{X}} \boldsymbol{\beta}_{\boldsymbol{x}} \left( \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} p(z \mid x,a,\mu_E)\nu(x,a) - \mu_E(z) \right) \\
& + \sum_{x\in\mathcal{X}} \boldsymbol{\theta}_{\boldsymbol{x}} \left(\nu^{\mathcal{X}}(x) - \mu_E(x)\right) \Bigg\}
\end{aligned}
$$

We can organize **(OPT)$_2$** as:

$$\textbf{(OPT)}_\mathbf{2} = \max_{\nu \in \mathcal{P}(\mathcal{X}\times\mathcal{A})} \min_{\substack{\boldsymbol{\alpha}\in\mathbb{R}^k \\ \boldsymbol{\beta},\boldsymbol{\theta}\in\mathbb{R}^{\mathcal{X}}}} \Bigg\{ H(\nu) + \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)\nu(x,a) \\ - \langle \boldsymbol{\alpha},\, \langle f\rangle_{\pi_E,\mu_E}\rangle - \sum_{x\in\mathcal{X}} \boldsymbol{\theta_x}\mu_E(x) \Bigg\},$$

where

$$k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a) := \log(\mu_E(x)) + \langle \boldsymbol{\alpha}, f(x,a,\mu_E)\rangle + \boldsymbol{\theta}_x + \sum_{z\in\mathcal{X}} \boldsymbol{\beta}_z\big(p(z \mid x,a,\mu_E) - \mu_E(z)\big).$$

Here, $H(\nu)$, as the entropy of $\nu$, is strongly concave in $\nu$, while the second term is linear in $\nu$. The function $k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)$ and the remaining terms are linear in $\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}$, guarantees the convexity of the objective in $\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}$.

**Theorem 1**

There are mappings between feasible points of $\mathbf{(OPT)_1}$ and $\mathbf{(OPT)_2}$, and under these mappings two equivalent feasible points lead to the same objective value.

*Proof.* ($\Rightarrow$) Let $\pi \in \Pi$ be feasible for $\mathbf{(OPT)_1}$. And let $\nu_\pi(x,a) := \mu_E(x)\,\pi(a \mid x)$. By Lemma 1 for all $x \in \mathcal{X}$ we have:

$$\mu_E(x) = \sum_{(y,a)\in\mathcal{X}\times\mathcal{A}} p(x \,|\, y, a, \mu_E)\nu_\pi(y,a).$$

Furthermore, from Lemma 3, we have:

$$\sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} f(x,a,\mu_E)\nu_\pi(x,a) = \langle f\rangle_{\pi_E,\mu_E}.$$

In addition, since $\nu_\pi$ is a probability distribution, we have the third and forth constraints. Since $\nu_\pi$ satisfies all the constraints $\mathbf{(OPT)_2}$ have, it is feasible for $\mathbf{(OPT)_2}$...

($\Leftarrow$) Let $\nu \in \mathcal{P}(\mathcal{X} \times \mathcal{A})$ be feasible for $\mathbf{(OPT)_2}$, with $\nu^{\mathcal{X}} \equiv \mu_E$. We define a policy $\pi_\nu \in \Pi$ by

$$\pi_\nu(a \mid x) := \frac{\nu(x,a)}{\mu_E(x)}.$$

where $\nu^{\mathcal{X}}(x) := \sum_{a \in \mathcal{A}} \nu(x,a)$. Moreover, it satisfies the stationarity condition

$$\mu_E(x) = \sum_{(y,a) \in \mathcal{X} \times \mathcal{A}} p(x \mid y, a, \mu_E)\, \nu(y,a), \quad \text{for all } x \in \mathcal{X}.$$

Using the definition of $\pi_\nu$, we may write $\nu(y,a) = \nu^{\mathcal{X}}(y)\, \pi_\nu(a \mid y) = \mu_E(y)\, \pi_\nu(a \mid y)$, and so the stationarity condition becomes

$$\mu_E(x) = \sum_{y,a} p(x \mid y, a, \mu_E)\, \pi_\nu(a \mid y)\, \mu_E(y),$$

which shows that $\mu_E \in \Lambda(\pi_\nu)$.

Now consider the occupation measure $\nu_{\pi_\nu} \in \mathcal{P}(\mathcal{X} \times \mathcal{A})$ defined under policy $\pi_\nu$ with invariant state distribution $\mu_E$, we have[3]:

$$\nu_{\pi_\nu}(x,a) = \mu_E(x)\,\pi_\nu(a \mid x) \implies \nu_{\pi_\nu}(x,a) = \mu_E(x)\,\frac{\nu(x,a)}{\nu^{\mathcal{X}}(x)} = \nu(x,a)$$

Whenever $\nu^{\mathcal{X}}(x) > 0$, and both sides are zero when $\nu^{\mathcal{X}}(x) = 0$ due to feasibility. Hence, $\nu_{\pi_\nu} \equiv \nu$, and in particular,

$$\nu^{\mathcal{X}}_{\pi_\nu}(x) = \sum_{a \in \mathcal{A}} \nu_{\pi_\nu}(x,a) = \mu_E(x), \quad \text{for all } x \in \mathcal{X}.$$

It remains to verify that $\pi_\nu \in \Pi$ is feasible for $\mathbf{(OPT)_1}$.

[3] *See the proof of Lemma 2*

We have already shown that $\mu_E \in \Lambda(\pi_\nu)$. Next, the expected feature constraint in **$(\mathbf{OPT})_1$** holds, since

$$\sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} f(x,a,\mu_E)\,\nu_{\pi_\nu}(x,a) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} f(x,a,\mu_E)\,\nu(x,a) = \langle f\rangle_{\pi_E,\mu_E},$$

by the feasibility of $\nu$ for **$(\mathbf{OPT})_2$** and Lemma 3. Moreover, by Lemma 2, the causal entropy of $\pi_\nu$ satisfies

$$H(\pi_\nu) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} -\log\left(\frac{\nu_{\pi_\nu}(x,a)}{\mu_E(x)}\right)\,\nu_{\pi_\nu}(x,a) = \sum_{(x,a)} -\log\left(\frac{\nu(x,a)}{\mu_E(x)}\right)\,\nu(x,a).$$

Thus, the policy $\pi_\nu \in \Pi$ is feasible for **$(\mathbf{OPT})_1$**, and it achieves the same objective value as $\nu$ in **$(\mathbf{OPT})_2$**. ■

# Remark

Let $\pi \in \Pi$ and $\nu \in \mathcal{P}(\mathcal{X} \times \mathcal{A})$, and suppose that $\mu_E(x) > 0$ for all $x \in \mathcal{X}$. Consider the following mappings between policies and occupation measures:

$$\pi \mapsto \nu_\pi := \mu_E(x)\, \pi(a \mid x) \in \mathcal{P}(\mathcal{X} \times \mathcal{A}),$$
$$\nu \mapsto \pi_\nu(a \mid x) := \frac{\nu(x,a)}{\nu^{\mathcal{X}}(x)} \in \Pi.$$

Under assumption $\nu^{\mathcal{X}}(x) = \mu_E(x) > 0$, the second mapping is well-defined. Moreover, the first mapping constructs the occupation measure induced by a policy $\pi$ and invariant distribution $\mu_E$, while the second mapping reconstructs the policy from a given occupation measure $\nu$.

# Dual of **(OPT)$_2$** and its Solution

### Theorem 2

The dual of **(OPT)$_2$** is

$$\min_{\substack{\boldsymbol{\alpha}\in\mathbb{R}^k\\ \boldsymbol{\beta},\boldsymbol{\theta}\in\mathbb{R}^{\mathcal{X}}}} \left\{ \log \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} e^{k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)} - \langle \boldsymbol{\alpha},\ \langle f\rangle_{\pi_E,\mu_E}\rangle - \sum_{x\in\mathcal{X}} \boldsymbol{\theta}_{\boldsymbol{x}}\mu_E(x) \right\}$$

where

$$k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a) := \log(\mu_E(x)) + \langle \boldsymbol{\alpha},\ f(x,a,\mu_E)\rangle + \boldsymbol{\theta}_{\boldsymbol{x}} + \sum_{z\in\mathcal{X}} \boldsymbol{\beta}_{\boldsymbol{z}}(p(z \mid x,a,\mu_E) - \mu_E(z)).$$

*Proof.* To achieve the dual form of **(OPT)$_2$**, we recall the Min-Max formulation of **(OPT)$_2$**. From Sion's minimax theorem[4], we can change the order of $\max$ and $\min$:

[4] M. Sion, "On general minimax theorems," *Pacific Journal of Mathematics*, vol. 8, pp. 171–176, 1958.

We have,

$$\mathbf{(OPT)_2} = \min_{\substack{\boldsymbol{\alpha}\in\mathbb{R}^k \\ \boldsymbol{\beta},\boldsymbol{\theta}\in\mathbb{R}^{\mathcal{X}}}} \left\{ \log \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} e^{k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)} - \langle \boldsymbol{\alpha},\, \langle f\rangle_{\pi_E,\mu_E}\rangle - \sum_{x\in\mathcal{X}} \boldsymbol{\theta}_{\boldsymbol{x}} \mu_E(x) \right\}$$

Here equality follows from the variational formula[5]

$$\log \sum_{t\in T} e^{k(t)} = \max_{\nu\in\mathcal{P}(T)} \left[ H(\nu) + \sum_{t\in T} k(t)\nu(t) \right].$$

Thus, above minimization problem is the dual of $\mathbf{(OPT)_2}$ and Sion's minimax theorem guarantees that there is no duality gap.

[5] P. Dupuis and R. S. Ellis, *Formulation of Large Deviation Theory in Terms of the Laplace Principle*, John Wiley & Sons, Ltd, 1997, ch. 1, pp. 1–47.

Also, for given $\boldsymbol{\alpha} \in \mathbb{R}^k$, $\boldsymbol{\beta}, \boldsymbol{\theta} \in \mathbb{R}^{\mathcal{X}}$, the probability measure $\nu^*$ that maximizes

$$H(\nu) + \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)\nu(x,a)$$

is the Boltzman distribution defined as:[6]

$$\nu^*(x,a) := \frac{e^{k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)}}{Z_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}},$$

where

$$Z_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}} := \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} e^{k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)}$$

■

[6] P. Dupuis and R. S. Ellis, *Formulation of Large Deviation Theory in Terms of the Laplace Principle*, John Wiley & Sons, Ltd, 1997, ch. 1, pp. 1–47.

# Dual Objective

From now on, we denote the objective function in the dual formulation of **(OPT)$_2$** as

$$h(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) := \log \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} e^{k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)} - \langle \boldsymbol{\alpha}, \langle f\rangle_{\pi_E,\mu_E}\rangle - \sum_{x\in\mathcal{X}} \boldsymbol{\theta}_{\boldsymbol{x}} \mu_E(x)$$

Note that the term $H(\nu) + \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)\, \nu(x,a)$ is linear in $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$. The maximum of this expression over $\nu$, which is given by

$$\log \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} e^{k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)},$$

is a convex function of $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$. Since the remaining terms in $h$ are linear in $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$, we can conclude that $h$ is a convex function.

**Theorem 3**

h is L-smooth where

$$L := 2M^2\sqrt{|\mathcal{X}| \cdot |\mathcal{A}|}$$

here M can be determined explicitly.

*Proof.* We begin by computing the partial gradients of the function $h$ with respect to the parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\theta}$ as follows:

$$\nabla_{\boldsymbol{\alpha}} h(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} \nu^*(x,a) f(x,a,\mu_E) - \langle f \rangle_{\pi_E,\mu_E},$$

$$\nabla_{\boldsymbol{\beta}} h(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} \nu^*(x,a)\, p(\,\cdot \mid x,a,\mu_E) - \mu_E(\,\cdot\,),$$

$$\nabla_{\boldsymbol{\theta}} h(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) = \sum_{(x,a)\in\mathcal{X}\times\mathcal{A}} \nu^*(x,a)\, e_x(\,\cdot\,) - \mu_E(\,\cdot\,),$$

where $e_x(y) := 1_{\{x=y\}}$.

To establish Lipschitz continuity (smoothness) of $\nabla h$, it suffices to show the Lipschitz continuity of $\nu^*$, as any discontinuity in $\nu^*$ would directly propagate to $\nabla h$. Thus, we compute the gradient of $\nu^*$ with respect to $\gamma \in \{\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}\}$:

$$\nabla_\gamma \nu^*_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x, a) = \nu^*(x, a) \left[\nabla_\gamma k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x, a) - \langle \nabla_\gamma k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}} \rangle_{\nu^*}\right].$$

Let us define the following uniform bounds over all $(x, a) \in \mathcal{X} \times \mathcal{A}$ and parameters $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}$:

$$\begin{aligned}
&\sup \|\nabla_{\boldsymbol{\alpha}} k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x, a)\| = \sup \|f(x, a, \mu_E)\| =: M_{\boldsymbol{\alpha}}, \\
&\sup \|\nabla_{\boldsymbol{\beta}} k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x, a)\| = \sup \|e_x\| =: M_{\boldsymbol{\beta}}, \\
&\sup \|\nabla_{\boldsymbol{\theta}} k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x, a)\| = \sup \|p(\cdot \mid x, a, \mu_E) - \mu_E(\cdot)\| =: M_{\boldsymbol{\theta}}
\end{aligned}$$

Define $M := \max\{M_{\boldsymbol{\alpha}}, M_{\boldsymbol{\beta}}, M_{\boldsymbol{\theta}}\}$. Therefore, for each $\gamma \in \{\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}\}$, we obtain:

$$\sup_{\substack{(x,a)\in\mathcal{X}\times\mathcal{A}\\ \boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}} \|\nabla_\gamma \nu^*(x,a)\| \ \le 2M.$$

By Mean Value Theorem, it follows that $\nu^*$ is $2M$-Lipschitz continuous with respect to parameters $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$.
Having established Lipschitz continuity of $\nabla_\gamma \nu^*$, we derive Lipschitz continuity of $\nabla h$, leading to:

$$\|\Delta(\nabla_\gamma h)\| \le M_\gamma\, 2M\, \|\Delta(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})\| \sqrt{|\mathcal{X}| \cdot |\mathcal{A}|},$$

where $\Delta(\cdot)$ denotes the difference between two evaluations at parameter points $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$ and $(\boldsymbol{\alpha}', \boldsymbol{\beta}', \boldsymbol{\theta}')$.

■

# Convexity

**Theorem 4**

Suppose that the set

$$\{(f(x,a,\mu_E),\ p(\cdot \mid x,a,\mu_E),\ e(\cdot \mid x,a)) : (x,a) \in \mathcal{X} \times \mathcal{A}\}$$

spans the space $\mathbb{R}^k \times \mathbb{R}^{\mathcal{X}} \times \mathbb{R}^{\mathcal{X}}$, where $e(y| x,a) := 1_{\{x=y\}}$. Then the function $h$ is $\rho(D)$-strongly convex over any compact subset $D \subset \mathbb{R}^m$, where $m = k + 2|\mathcal{X}|$.

*Proof.* Our aim is to establish the strong convexity of the function $h$. To demonstrate this, it suffices to show that the Hessian of $h$ is uniformly positive definite over compact subsets of the parameter space. [7]

[7] Boyd, S., Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press, Section 9.1.2.

We compute the partial Hessians as follows:

$$\nabla^2_{\boldsymbol{\alpha},\boldsymbol{\alpha}} h(.) = \mathsf{Cov}_{(x,a)\sim\nu^*}\Big(f(x,a,\mu_E)\Big)$$

$$\nabla^2_{\boldsymbol{\beta},\boldsymbol{\beta}} h(.) = \mathrm{Cov}_{(x,a)\sim\nu^*}\Big(p(\cdot|x,a,\mu_E)\Big).$$

$$\nabla^2_{\boldsymbol{\theta},\boldsymbol{\theta}} h(.) = \mathrm{Cov}_{(x,a)\sim\nu^*}\big(e(\cdot|x,a)\big).$$

Similarly, cross terms computed as:

$$\nabla^2_{\boldsymbol{\alpha},\boldsymbol{\beta}} h(.) = \mathrm{Cov}_{(x,a)\sim\nu^*}\big(p(\cdot|x,a,\mu_E), f(x,a,\mu_E)\big).$$

$$\nabla^2_{\boldsymbol{\alpha},\boldsymbol{\theta}} h(.) = \mathrm{Cov}_{(x,a)\sim\nu^*}\big(e(\cdot|x,a), f(x,a,\mu_E)\big).$$

$$\nabla^2_{\boldsymbol{\beta},\boldsymbol{\theta}} h(.) = \mathrm{Cov}_{(x,a)\sim\nu^*}\big(e(\cdot|x,a), p(\cdot|x,a,\mu_E)\big).$$

we define the random vector

$$\boldsymbol{X}(x,a) := (f(x,a,\mu_E),\ p(\,\cdot \,|\, x,a,\mu_E),\ e(\cdot \,|\, x,a))$$

Then we have $\mathrm{Hes}(h)(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}) = \mathrm{Cov}(\boldsymbol{X})$.
Assume, for the sake of contradiction, that $\mathrm{Cov}(\boldsymbol{X})$ is not positive definite. This means, by definition, there exists a nonzero vector $\boldsymbol{v} \in \mathbb{R}^m$ such that:

$$\langle \boldsymbol{v}, \mathrm{Cov}(\boldsymbol{X})\boldsymbol{v}\rangle = 0 \implies \sum_{i,j=1}^{m} v_j \,\mathrm{Cov}(\boldsymbol{X}_j, \boldsymbol{X}_i) v_i = \mathrm{Var}\left(\sum_{i=1}^{m} v_i \boldsymbol{X}_i\right) = 0$$

which implies that the random variable $\sum_{i=1}^{m} v_i \boldsymbol{X}_i$ is almost surely constant. This immediately implies that the support of $\mathrm{Law}(\boldsymbol{X})$ lies entirely within a hyperplane

$$\mathrm{supp}(\mathrm{Law}(\boldsymbol{X})) \subseteq \{\boldsymbol{d} \in \mathbb{R}^m : \langle \boldsymbol{v}, \boldsymbol{d}\rangle = c\}.$$

Since $\boldsymbol{v}$ is nonzero, this hyperplane is a lower-dimensional affine subspace of dimension at most $m-1$.

However, since $\boldsymbol{X}$ is defined as the mapping

$$\boldsymbol{X}(x,a) = \left(f(x,a,\mu_E),\ p(\,\cdot\,|\,x,a,\mu_E),\ e(\cdot\,|\,x,a)\right),$$

which transforms the state-action space $\mathcal{X} \times \mathcal{A}$ into $\mathbb{R}^m$. The probability distribution of $\boldsymbol{X}$ is the push-forward measure of $\nu^*$. That is, for any measurable set $A \subseteq \mathbb{R}^m$,

$$(\mathrm{Law}(\boldsymbol{X}) =)\, \mathbb{P}(\boldsymbol{X} \in A) = \nu^*(\{(x,a) \in \mathcal{X} \times \mathcal{A} \mid \boldsymbol{X}(x,a) \in A\}).$$

Since the Boltzmann policy $\nu^* \propto \exp(k_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}})$ assigns positive probability to every state-action pair in $\mathcal{X} \times \mathcal{A}$, the push-forward measure is strictly positive on its support. Furthermore, by Assumption,

$$\mathrm{span}\left\{\,(f(x,a,\mu_E),\ p(\,\cdot\,|\,x,a,\mu_E),\ e(\cdot\,|\,x,a)) : (x,a) \in \mathcal{X} \times \mathcal{A}\right\} = \mathbb{R}^m.$$

This contradicts the previous conclusion that $\mathrm{supp}(\mathrm{Law}(\boldsymbol{X}))$ is contained to a lower-dimensional hyperplane. Therefore, $\mathrm{Cov}(\boldsymbol{X})$ must be positive definite. This implies that $\mathrm{Law}(\boldsymbol{X})$ cannot be confined to a lower-dimensional subset of $\mathbb{R}^m$.

Now, consider the dependence of $\mathrm{Cov}(\boldsymbol{X})$ on the parameters $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$. Since $\boldsymbol{X}$ is a function of $(x, a) \sim \nu^*$, the covariance matrix $\mathrm{Cov}(\boldsymbol{X})$ continuously varies with respect to these parameters. Consequently, the smallest eigenvalue function $\lambda_{\min}(\boldsymbol{X})$ is also a continuous function of $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$.

Next, consider a compact subset $D \subset \mathbb{R}^m$ of the parameter space. By the Extreme Value Theorem, we know that $\lambda_{\min}(\boldsymbol{X})$ reaches some minimum value over $D$. Define

$$\min_{(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) \in D} \lambda_{\min}(\boldsymbol{X}) =: \lambda_{\min}(D) > 0.$$

Since Cov($\boldsymbol{X}$) is symmetric $m \times m$ matrix, we use its eigenvalue decomposition for all $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) \in D$ we have Cov$(\boldsymbol{X}) = U \Lambda U^T$ where $U$ is an orthogonal matrix containing the eigenvectors and $\Lambda$ is a diagonal matrix with eigenvalues on diagonal.
Since $\lambda_i \geq \lambda_{min}(D)$ for all $i \in \{1, 2, \ldots, m\}$, we can write;

$$\begin{aligned} \Lambda \succcurlyeq \lambda_{min}(D) I_m &\implies U \Lambda U^T \succcurlyeq U \lambda_{min}(D) I_m U^T \\ &\iff \text{Cov}(\boldsymbol{X}) = \mathrm{Hes}(h) \succcurlyeq \lambda_{\min}(D) I_m \end{aligned}$$

Hence, $h$ is $\rho(D)$-strongly convex on $D$, where

$$\rho(D) := \lambda_{\min}(D).$$

■

# Gradient Decent Algorithm

Now we can introduce the gradient descent algorithm for finding the minimizer of $h$ as follows.

**Algorithm 1** Gradient Descent for Parameter Optimization

**Require:** Initial parameters $(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0, \boldsymbol{\theta}_0)$, step size $\delta > 0$, number of iterations $K$

Set $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) \leftarrow (\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0, \boldsymbol{\theta}_0)$
**for** $k = 0, 1, \ldots, K-1$ **do**
    $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) \leftarrow (\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta}) - \delta \nabla h(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$
**end for**
Compute $\nu^*_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}$ from the final parameters.
**return** $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$ and $\nu^*_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}$

# Is Gradient Decent Valid?

Suppose that the step-size in gradient descent algorithm satisfies $0 < \delta \leq \frac{1}{L}$. Define the following compact subset of $\mathbb{R}^m$:

$$D := \left\{ \boldsymbol{d} \in \mathbb{R}^m : \|\boldsymbol{d} - (\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*)\| \leq \|(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0, \boldsymbol{\theta}_0) - (\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*)\| \right\}.$$

Then, for any $k$, we have

$$\begin{aligned} &\|(\boldsymbol{\alpha}_k, \boldsymbol{\beta}_k, \boldsymbol{\theta}_k) - (\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*)\| \\ &\leq (1 - \delta\, \rho(D))^k \, \|(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0, \boldsymbol{\theta}_0) - (\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*)\|. \end{aligned}$$

Therefore, since $\nu^*_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\theta}}(x,a)$ is $2M$-Lipschitz continuous with respect to $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\theta})$ for all $(x,a) \in \mathcal{X} \times \mathcal{A}$, we also have

$$\begin{aligned} &\|\nu^*_{\boldsymbol{\alpha}_k,\boldsymbol{\beta}_k,\boldsymbol{\theta}_k} - \nu^*_{\boldsymbol{\alpha}_*,\boldsymbol{\beta}_*,\boldsymbol{\theta}_*}\| \\ &\leq \sqrt{|\mathcal{X}||\mathcal{A}|}\, 2M \, (1 - \delta\, \rho(D))^k \, \|(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0, \boldsymbol{\theta}_0) - (\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*)\|. \end{aligned}$$

# Conclusion

If $(\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*)$ is the minimizer of $h$, which can be determined using the gradient descent algorithm, then the optimal solution to $\mathbf{(OPT)_1}$ is given by:

$$\nu^*_{\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*}(x, a) = \frac{e^{k_{\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*}(x,a)}}{\sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} e^{k_{\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*}(x,a)}}.$$

The policy corresponding to this distribution is:

$$\pi_{\nu^*_{\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*}}(a|x) = \frac{\nu^*_{\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*}(x, a)}{\nu^{*,\mathcal{X}}_{\boldsymbol{\alpha}_*, \boldsymbol{\beta}_*, \boldsymbol{\theta}_*}(x)}.$$

This policy solves the maximum causal entropy problem $\mathbf{(OPT)_1}$.

# Numerical Illustration

We illustrate our methodology using a malware spread model[8] under an average-cost criterion. Agents possess local states $x(t) \in \{0,1\}$, representing healthy $(0)$ or infected $(1)$ conditions. At each step, agents select actions $a(t) \in \{0,1\}$, where $0$ corresponds to "do nothing" and $1$ to "repair." The system dynamics follow

$$x(t+1) = \begin{cases} x(t) + (1 - x(t))w(t), & \text{if } a(t) = 0, \\ 0, & \text{if } a(t) = 1, \end{cases}$$

where $w(t) \sim \text{Bernoulli}(q = 0.9)$. Thus, a healthy agent that does nothing becomes infected with probability $q$, whereas repair guarantees recovery.

[8] J. Subramanian and A. Mahajan, "Reinforcement learning in stationary mean-field games," *Proceedings of the International Foundation for Autonomous Agents and Multiagent Systems*, 2019, pp. 251–259.

# Numerical Illustration

The per-stage cost is given by

$$c(x, a, \mu) = \theta_1 x + \theta_2 x \mu(1) + \theta_3 a.$$

Here, $\mu(1)$ denotes the fraction of infected agents, $\theta_3$ the repair cost, and $\theta_1 + \theta_2\mu(1)$ the risk associated with infection.
In the IRL problem, the variables $(\theta_1, \theta_2, \theta_3)$ are unknown. We use the linearity assumption:

$$\mathcal{R} := \left\{ c(x, a, \mu) = \langle \theta, f(x, a, \mu) \rangle \;\middle|\; \theta \in \mathbb{R}^3,\; f : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \to \mathbb{R}^3 \right\}$$

where $f(x, a, \mu(1)) = (x, x\mu(1), a)$.

Our goal is to recover $(\pi_E, \mu_E)$ using IRL.
We first consider the forward problem with known parameters $\theta = [0.3, 0.5, 0.8]$ and obtain the mean-field equilibrium $(\pi_E, \mu_E)$ numerically as:

$$\pi_E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mu_E \approx \begin{bmatrix} 0.52 \\ 0.47 \end{bmatrix}.$$

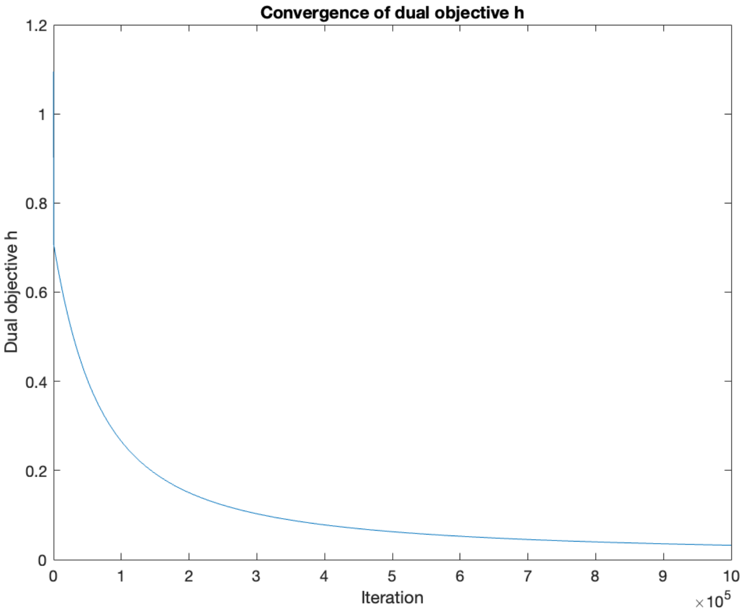
This equilibrium indicates that healthy agents do nothing, whereas infected agents always choose repair.

# Results

Using the MFE obtained numeratically, we solve the dual IRL problem to recover the optimal joint distribution $\nu(x, a)$, from which the policy was obtained via $\pi(a \mid x) = \frac{\nu(x,a)}{\mu_E(x)}$.
The resulting policy is

$$\pi = \begin{bmatrix} 0.9709 & 0.0291 \\ 0.0303 & 0.9697 \end{bmatrix} \quad \text{versus} \quad \pi_E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The recovered strategy closely approximated the expert behavior. The convergence of the dual objective $h$ is illustrated in Fig. 1, confirming the numerical consistency of the algorithm.


Convergence of dual objective h
Dual objective h
Iteration
$\times 10^5$
0
0.2
0.4
0.6
0.8
1
1.2
0
1
2
3
4
5
6
7
8
9
10